



MS/MA IN ECONOMICS

DYNAMIC STOCHASTIC MODELS

Lecture 01

Solow Model

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1 Solow Model with a Capital Externality

1.1 Firm's problem and aggregation

Each firm j operates the production technology

$$y_t^j = A_t k_t^\theta h_t^{1-\theta} K_t^{1-\theta}, \quad (1)$$

where lower-case letters (k_t, h_t) denote the firm's own capital and labor, while the upper-case K_t denotes the economy-wide capital stock. Each firm takes K_t as given: the aggregate capital stock is an *externality*, not internalized in the firm's optimization.

Private returns to scale. In its own factors, the firm exhibits constant returns to scale, since the exponents on k_t and h_t sum to $\theta + (1 - \theta) = 1$. Hence factors can be paid their marginal products under perfect competition.

We derive the equilibrium. Factor markets are competitive: every firm rents capital at the common rental rate r_t and hires labor at the common wage w_t , both taken as given. Since the Solow model contains no intertemporal saving decision, each firm solves a *static* profit-maximization problem period by period, taking the aggregate stock K_t as given (the externality):

$$\max_{k_t^j, h_t^j} A_t (k_t^j)^\theta (h_t^j)^{1-\theta} K_t^{1-\theta} - r_t k_t^j - w_t h_t^j. \quad (2)$$

The first-order conditions are

$$\theta A_t (k_t^j)^{\theta-1} (h_t^j)^{1-\theta} K_t^{1-\theta} = r_t, \quad (3)$$

$$(1 - \theta) A_t (k_t^j)^\theta (h_t^j)^{-\theta} K_t^{1-\theta} = w_t. \quad (4)$$

Symmetry as a result. Dividing (3) by (4) gives the firm's capital intensity,

$$\frac{k_t^j}{h_t^j} = \frac{\theta}{1 - \theta} \frac{w_t}{r_t}. \quad (5)$$

This ratio depends only on the parameter θ and on the economy-wide prices (r_t, w_t) , which are common across firms. Hence every firm chooses the *same* capital intensity. Together with market clearing, $\sum_{j=1}^J k_t^j = K_t$ and $\sum_{j=1}^J h_t^j = N_t$, this implies $k_t^j = k_t = K_t/J$ and $h_t^j = h_t = N_t/J$ for every firm. Symmetry is therefore an *equilibrium outcome*. (Equivalently: with constant private returns the firm's scale is indeterminate, so the symmetric allocation is without loss of generality.)

Aggregate technology. Substituting the symmetric allocation into (1) and aggregating yields ¹

$$Y_t = A_t K_t^\theta N_t^{1-\theta} K_t^{1-\theta} = A_t K_t N_t^{1-\theta}. \quad (6)$$

Note that each firm pays capital its *private* marginal product (governed by the exponent θ), while the *social* return to aggregate capital is governed by the unit exponent, removing the diminishing returns to capital that characterize the standard Solow model. This wedge between the private return (θ) and the social return (1) is the source of the externality and of the gap between the decentralized equilibrium and the planner's allocation. It alters the model's growth and convergence properties.

¹See Anex

1.2 Law of motion with constant population ($\lambda = 0$)

Set $\lambda = 0$, so population is constant, $N_t = N$, and take constant technology $A_t = A$. Combining the savings–investment equilibrium $S_t = I_t = sY_t$ with the accumulation rule $K_{t+1} = (1 - \delta)K_t + I_t$ and the aggregate technology (6),

$$K_{t+1} = (1 - \delta)K_t + sA K_t N^{1-\theta}. \quad (7)$$

Dividing by N and writing $k_t = K_t/N$,

$$k_{t+1} = (1 - \delta)k_t + sAN^{1-\theta}k_t. \quad (8)$$

Defining $\bar{A} \equiv AN^{1-\theta}$, the law of motion collapses to the *linear* difference equation

$$k_{t+1} = [(1 - \delta) + s\bar{A}] k_t. \quad (9)$$

An AK model. The externality has turned the concave Solow term k_t^θ into a linear term in k_t . Comparing,

$$\text{Standard Solow: } k_{t+1} = (1 - \delta)k_t + sA_0 k_t^\theta, \quad \theta < 1, \quad (10)$$

$$\text{With externality: } k_{t+1} = (1 - \delta)k_t + s\bar{A} k_t, \quad (11)$$

so diminishing returns to capital have been eliminated.

Growth rate. Dividing (9) by k_t gives the gross growth rate of per-worker capital,

$$\gamma_t \equiv \frac{k_{t+1}}{k_t} = (1 - \delta) + s\bar{A}, \quad (12)$$

which is *constant*, independent of k_t . The net growth rate is

$$\gamma_t - 1 = s\bar{A} - \delta. \quad (13)$$

Unlike the standard Solow model—where the growth rate falls with k_t and generates convergence (poor economies grow faster)—here the growth rate is independent of the capital stock. There is *no convergence*, and the economy displays sustained (endogenous) growth whenever $s\bar{A} > \delta$, with no need for exogenous technological progress.

1.3 Explicit time path

Since (9) is linear with constant gross growth factor $\gamma \equiv (1 - \delta) + s\bar{A}$, the law of motion is a geometric progression. Iterating from an initial k_0 ,

$$k_t = \gamma^t k_0 = [(1 - \delta) + s\bar{A}]^t k_0. \quad (14)$$

This is the *exact* time path; unlike the standard Solow model, no numerical iteration or approximation is required. Output per worker inherits the same path, since $y_t = \bar{A}k_t = \bar{A}\gamma^t k_0$, so capital and output grow at the common constant rate γ , as on a balanced growth path. The behavior depends entirely on γ :

- $\gamma > 1$ ($s\bar{A} > \delta$): k_t grows without bound (sustained endogenous growth);
- $\gamma = 1$ ($s\bar{A} = \delta$): $k_t = k_0$ for all t (knife-edge);
- $\gamma < 1$ ($s\bar{A} < \delta$): $k_t \rightarrow 0$ (the economy collapses).

1.4 Stationary states and stability

Stationary states in *levels* solve $k = \gamma k$, i.e. $k(1 - \gamma) = 0$. For $\gamma \neq 1$ the only solution is $k = 0$: the interior attractor of the standard Solow model, $k^* = (sA_0/(\delta + \lambda))^{1/(1-\theta)}$, has disappeared. Geometrically, $G(k) = \gamma k$ is a straight line through the origin with slope γ , which can only meet the 45° line at the origin (unless $\gamma = 1$, when the two lines coincide). Stability of $k = 0$ follows from the slope:

- $\gamma > 1$: the line lies above the diagonal; $k = 0$ is *unstable* (any $k_0 > 0$ diverges);
- $\gamma < 1$: the line lies below the diagonal; $k = 0$ is *stable* (every path collapses to the origin);
- $\gamma = 1$: degenerate case, every k_0 is stationary.

In contrast to the standard Solow model, the linear dynamics admit no finite long-run level of capital. The relevant long-run object is no longer a level to which the economy converges, but a constant growth *rate*—the defining feature of endogenous growth.

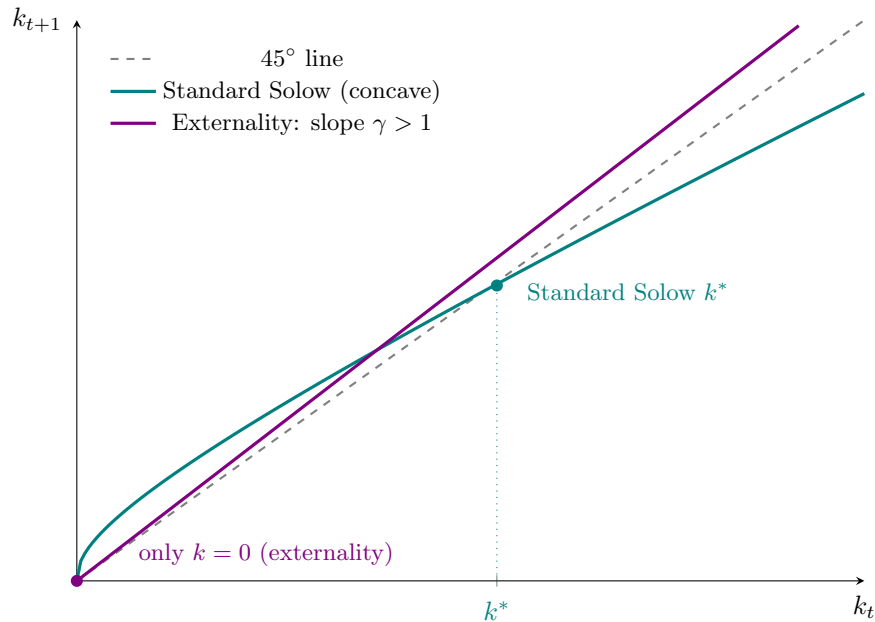


Figure 1: Phase diagram. The standard Solow law of motion (teal) is concave: it lies above the 45° line for $k_t < k^*$ and below it for $k_t > k^*$, crossing once from above at the stable interior steady state k^* . With the capital externality the law of motion is linear, $k_{t+1} = \gamma k_t$ (violet); for $\gamma > 1$ it lies everywhere above the 45° line and meets it only at the origin. The interior steady state k^* belongs to the standard model *only*: the externality eliminates it, leaving $k = 0$ as the unique (repelling) steady state, so the economy grows without bound.

1.5 Stochastic version

Following the standard treatment, let technology be stochastic, $A_t = A e^{\varepsilon_t}$, with $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$. The effective scale constant becomes $\bar{A}_t \equiv A_t N^{1-\theta} = \bar{A} e^{\varepsilon_t}$, with $\bar{A} = A N^{1-\theta}$. The stochastic law of motion is

$$k_{t+1} = [(1 - \delta) + s \bar{A} e^{\varepsilon_t}] k_t, \quad (15)$$

with stochastic gross growth rate

$$\gamma_t = (1 - \delta) + s \bar{A} e^{\varepsilon_t}. \quad (16)$$

Unlike the standard Solow model, where the shock multiplies the concave term k_t^θ (so diminishing returns pull the economy back toward its stationary state), here the shock multiplies k_t directly and no such mean-reversion operates.

1.6 Log-linear version and persistence

Taking logs of the growth rate, the standard Solow model yields (cf. the lecture)

$$\ln\left[\gamma_t - \frac{1-\delta}{1+\lambda}\right] = \varphi - (1-\theta) \ln k_t + \varepsilon_t, \quad (17)$$

where the term $-(1-\theta) \ln k_t$ drives convergence. In the present model the social exponent on capital is unity, so $1-\theta$ is replaced by $1-1=0$ and this term vanishes:

$$\ln[\gamma_t - (1-\delta)] = \ln(s\bar{A}) + \varepsilon_t. \quad (18)$$

The growth rate of capital responds only to the contemporaneous shock, not to the inherited capital stock.

Log-linearized capital. Since no finite stationary state exists when $\gamma \neq 1$, we linearize around the knife-edge $s\bar{A} = \delta$ (where $\gamma = 1$ and a stationary level is well defined), to compare cleanly with the standard model. With $\tilde{k}_t \equiv \ln(k_t/\bar{k})$ and $e^{\varepsilon_t} \approx 1 + \varepsilon_t$, equation (15) becomes, to first order,

$$\tilde{k}_{t+1} = \tilde{k}_t + \delta \varepsilon_t. \quad (19)$$

This is a *random walk*. Compare with the standard Solow result

$$\tilde{k}_{t+1} = B \tilde{k}_t + C \varepsilon_t, \quad B = \frac{1 + \lambda\theta - \delta(1-\theta)}{1+\lambda} < 1, \quad (20)$$

whose solution $\tilde{k}_{t+1} = C \sum_{i \geq 0} B^i \varepsilon_{t-i}$ is convergent with finite variance $\frac{C^2}{1-B^2} \sigma^2$. Here the analogous coefficient is $B = 1$: the externality has removed diminishing returns, so

$$\tilde{k}_{t+1} = \sum_{i=0}^{\infty} \delta \varepsilon_{t-i} \quad (21)$$

does not converge, and the variance of capital grows without bound.

Interpretation. The standard Solow model has $B < 1$ and therefore cannot generate persistence of cycles. The capital externality drives $B \rightarrow 1$, producing *maximal* persistence: a technology shock shifts capital permanently. The model overcorrects the Solow shortcoming, moving from no persistence to a unit root. This is why RBC models require shock processes of *intermediate* persistence (an AR(1) with $|\gamma| < 1$): neither pure Solow nor pure AK matches the data.

2 Quadratic Capital Adjustment Costs

2.1 Setup: the resource constraint

In the Solow model there is no intertemporal optimization, so adjustment costs cannot enter through an optimized investment decision. Instead they enter as a *resource cost*: changing the capital stock absorbs output. The resource constraint becomes

$$Y_t = C_t + I_t + \phi(K_{t+1} - K_t)^2, \quad (22)$$

so that, with the Solow savings rule, investment available for accumulation is $I_t = sY_t - \phi(K_{t+1} - K_t)^2$. With $\lambda = 0$, constant population N , the externality technology $Y_t = AK_t N^{1-\theta}$, and $\bar{A} \equiv AN^{1-\theta}$, the accumulation rule is

$$K_{t+1} = (1-\delta)K_t + s\bar{A}K_t - \phi(K_{t+1} - K_t)^2. \quad (23)$$

Setting $\phi = 0$ recovers the linear AK law of motion $K_{t+1} = [(1 - \delta) + s\bar{A}]K_t$ derived earlier.

2.2 Solving the implicit equation

Equation (23) is implicit in K_{t+1} , which appears quadratically. Writing $\Delta K_{t+1} \equiv K_{t+1} - K_t$ and subtracting K_t from both sides,

$$\phi (\Delta K_{t+1})^2 + \Delta K_{t+1} - (s\bar{A} - \delta)K_t = 0. \quad (24)$$

Solving this quadratic and taking the economically relevant (positive) root,

$$\Delta K_{t+1} = \frac{-1 + \sqrt{1 + 4\phi(s\bar{A} - \delta)K_t}}{2\phi}, \quad (25)$$

so the law of motion is

$$K_{t+1} = K_t + \frac{-1 + \sqrt{1 + 4\phi(s\bar{A} - \delta)K_t}}{2\phi}. \quad (26)$$

As $\phi \rightarrow 0$, a first-order expansion $\sqrt{1 + \epsilon} \approx 1 + \epsilon/2$ gives $\Delta K_{t+1} \rightarrow (s\bar{A} - \delta)K_t$, recovering the AK model.

Concavity. The adjustment cost replaces the *linear* AK law of motion with one containing a *concave* square root in K_t . By penalizing large changes in the capital stock, the quadratic cost reintroduces curvature into the dynamics and dampens the explosive growth of the pure AK model—an effect analyzed next.

2.3 Growth rate and long-run behavior

Dividing the change in capital by K_t , the net growth rate is

$$\gamma_t - 1 = \frac{1}{K_t} \frac{-1 + \sqrt{1 + 4\phi(s\bar{A} - \delta)K_t}}{2\phi}. \quad (27)$$

Unlike the pure AK model, where $\gamma - 1 = s\bar{A} - \delta$ is constant, the growth rate now depends on K_t . The limiting behavior is informative.

Small K_t . Expanding the root for small $\epsilon = 4\phi(s\bar{A} - \delta)K_t$,

$$\Delta K_{t+1} \approx (s\bar{A} - \delta)K_t, \quad \gamma_t - 1 \approx s\bar{A} - \delta, \quad (28)$$

so the economy initially grows almost as in the AK model: adjustment costs are negligible when capital changes are small.

Large K_t . The constant under the root becomes negligible, giving

$$\Delta K_{t+1} \approx \sqrt{\frac{s\bar{A} - \delta}{\phi}} \sqrt{K_t}, \quad \gamma_t - 1 \approx \sqrt{\frac{s\bar{A} - \delta}{\phi}} \frac{1}{\sqrt{K_t}} \xrightarrow{K_t \rightarrow \infty} 0. \quad (29)$$

The growth rate *declines* with K_t and tends to zero. The quadratic adjustment cost restores the convergence property that the externality had removed:

- Standard Solow: γ_t falls with k_t (convergence);
- Externality (AK): γ_t constant (no convergence);
- Externality + adjustment cost: γ_t falls with K_t (convergence restored).

Long-run path. Treating the large- K approximation as a continuous law $\dot{K} \approx C\sqrt{K}$ with $C = \sqrt{(s\bar{A} - \delta)/\phi}$ and integrating,

$$2\sqrt{K} = Ct + \text{const} \implies K_t \propto t^2. \quad (30)$$

Capital grows *quadratically* in time rather than exponentially: the adjustment cost converts the explosive exponential growth of the AK model into moderate polynomial growth. The economy still grows without bound, but at an ever-decreasing rate.

2.4 Log-linear version

With stochastic technology $A_t = Ae^{\varepsilon_t}$, the implicit law of motion is

$$K_{t+1} = (1 - \delta)K_t + s\bar{A}e^{\varepsilon_t}K_t - \phi(K_{t+1} - K_t)^2. \quad (31)$$

We linearize around the knife-edge $s\bar{A} = \delta$, where an interior steady state $\bar{K}$ with $\Delta K = 0$ is well defined (as for the AK model). Let $\tilde{K}_t \equiv \ln(K_t/\bar{K})$, so $K_t \approx \bar{K}(1 + \tilde{K}_t)$.

The adjustment term is second order. In deviations,

$$K_{t+1} - K_t = \bar{K}(\tilde{K}_{t+1} - \tilde{K}_t), \quad \phi(K_{t+1} - K_t)^2 = \phi\bar{K}^2(\tilde{K}_{t+1} - \tilde{K}_t)^2. \quad (32)$$

Since $(\tilde{K}_{t+1} - \tilde{K}_t)$ is first order, its square is second order and drops out of a first-order approximation:

$$\phi\bar{K}^2(\tilde{K}_{t+1} - \tilde{K}_t)^2 \approx 0. \quad (33)$$

Hence, to first order, the quadratic adjustment cost *vanishes* from the log-linearized dynamics. Linearizing the remaining terms exactly as in the AK case ($e^{\varepsilon_t} \approx 1 + \varepsilon_t$, $s\bar{A} = \delta$),

$$\tilde{K}_{t+1} = \tilde{K}_t + \delta\varepsilon_t, \quad (34)$$

which is the same random walk obtained for the pure AK model: the persistence coefficient remains $B = 1$.

Interpretation: two regimes. The quadratic adjustment cost acts differently on global and local dynamics:

- *Far from the steady state* (large levels): the cost matters, slowing growth from exponential to quadratic and restoring convergence;
- *Near the steady state* (small fluctuations): the cost is second order and disappears from the linear approximation, leaving the shock persistence unchanged at $B = 1$.

Thus the quadratic cost disciplines long-run growth but does *not* help with the short-run persistence of cycles discussed earlier. Addressing that would require an adjustment cost with a first-order effect.

3 Bonus Track: Linear (Non-Quadratic) Adjustment Costs

3.1 Setup and law of motion

Now the adjustment cost is linear in the change of capital, $\phi(K_{t+1} - K_t)$. The resource constraint is

$$Y_t = C_t + I_t + \phi(K_{t+1} - K_t), \quad (35)$$

so investment available for accumulation is $I_t = sY_t - \phi(K_{t+1} - K_t)$. With $\lambda = 0$ and $Y_t = \bar{A}K_t$,

$$K_{t+1} = (1 - \delta)K_t + s\bar{A}K_t - \phi(K_{t+1} - K_t). \quad (36)$$

Because the cost is *linear* in K_{t+1} , the equation is implicit but linear and can be solved explicitly—no quadratic formula or root selection is needed. Collecting terms in K_{t+1} ,

$$K_{t+1}(1 + \phi) = [(1 - \delta) + s\bar{A} + \phi]K_t, \quad (37)$$

$$K_{t+1} = \frac{(1 - \delta) + s\bar{A} + \phi}{1 + \phi} K_t. \quad (38)$$

3.2 Growth rate: a convex combination

The law of motion (38) is *again linear* in K_t : the linear cost does not introduce concavity (unlike the quadratic case). The gross growth factor is

$$\gamma^\phi \equiv \frac{(1 - \delta) + s\bar{A} + \phi}{1 + \phi} = \frac{\gamma + \phi}{1 + \phi} = \frac{1}{1 + \phi} \gamma + \frac{\phi}{1 + \phi} \cdot 1, \quad (39)$$

where $\gamma = (1 - \delta) + s\bar{A}$ is the pure- AK factor. Thus γ^ϕ is a convex combination of γ and 1:

- $\phi = 0$: $\gamma^\phi = \gamma$ (pure AK recovered);
- $\phi \rightarrow \infty$: $\gamma^\phi \rightarrow 1$ (capital frozen: adjusting is infinitely costly);
- intermediate ϕ : γ^ϕ lies between 1 and γ , so for $\gamma > 1$ the cost *lowers* the growth rate toward 1 while keeping it constant.

Contrast with the quadratic case. The two specifications behave qualitatively differently:

- Quadratic cost: growth becomes *decreasing* in K_t (convergence, $K_t \propto t^2$); large changes are penalized disproportionately.
- Linear cost: growth remains *constant* (exponential, no convergence), merely slower; the cost acts as a flat “tax” on capital changes.

The linear cost preserves the AK structure; the quadratic cost breaks it.

3.3 Time path

Since (38) is linear with constant factor γ^ϕ ,

$$K_t = (\gamma^\phi)^t K_0 = \left(\frac{(1 - \delta) + s\bar{A} + \phi}{1 + \phi} \right)^t K_0. \quad (40)$$

Growth is exponential, as in the pure AK model, but with base $\gamma^\phi < \gamma$. The steady-state and stability analysis is identical in structure to the AK case (Section on the externality model) with γ replaced by γ^ϕ : the unique interior steady state is $K = 0$, which repels if $\gamma^\phi > 1$. The sustained-growth condition is

$$\gamma^\phi > 1 \iff s\bar{A} > \delta, \quad (41)$$

the *same* as without the cost: the ϕ cancels, so the linear cost slows growth but does not determine whether the economy grows.

3.4 Log-linear version

With $\bar{A}_t = \bar{A}e^{\varepsilon_t}$, the stochastic implicit equation is

$$K_{t+1}(1 + \phi) = [(1 - \delta) + \phi]K_t + s\bar{A}e^{\varepsilon_t}K_t. \quad (42)$$

The deterministic steady state again requires $s\bar{A} = \delta$ (the ϕ cancels). Let $\tilde{K}_t = \ln(K_t/\bar{K})$. Crucially, the cost is now *linear* in $K_{t+1} - K_t = \bar{K}(\tilde{K}_{t+1} - \tilde{K}_t)$, a *first-order* deviation, so—unlike the quadratic case—it survives the linearization. Expanding to first order with $e^{\varepsilon_t} \approx 1 + \varepsilon_t$ and $s\bar{A} = \delta$, the constant $(1 + \phi)$ terms cancel and

$$(1 + \phi)\tilde{K}_{t+1} = (1 + \phi)\tilde{K}_t + \delta\varepsilon_t \implies \tilde{K}_{t+1} = \tilde{K}_t + \frac{\delta}{1 + \phi}\varepsilon_t. \quad (43)$$

Comparison of the three models.

Model	Log-linear law	Persistence B	Shock loading
Pure AK	$\tilde{k}_{t+1} = \tilde{k}_t + \delta\varepsilon_t$	1	δ
Quadratic cost	$\tilde{K}_{t+1} = \tilde{K}_t + \delta\varepsilon_t$	1	δ
Linear cost	$\tilde{K}_{t+1} = \tilde{K}_t + \frac{\delta}{1 + \phi}\varepsilon_t$	1	$\frac{\delta}{1 + \phi}$

The linear cost attenuates the *loading* on the shock by $1/(1 + \phi)$: a given technology shock moves capital less, because adjusting is costly. The *persistence*, however, remains $B = 1$ in all three cases. Neither adjustment-cost specification resolves the short-run persistence problem (all remain random walks): the quadratic cost reshapes long-run growth (exponential $\rightarrow t^2$), while the linear cost dampens the amplitude of short-run fluctuations. Generating intermediate persistence of cycles requires persistence in the shock process itself (an AR(1)), not in the accumulation technology.

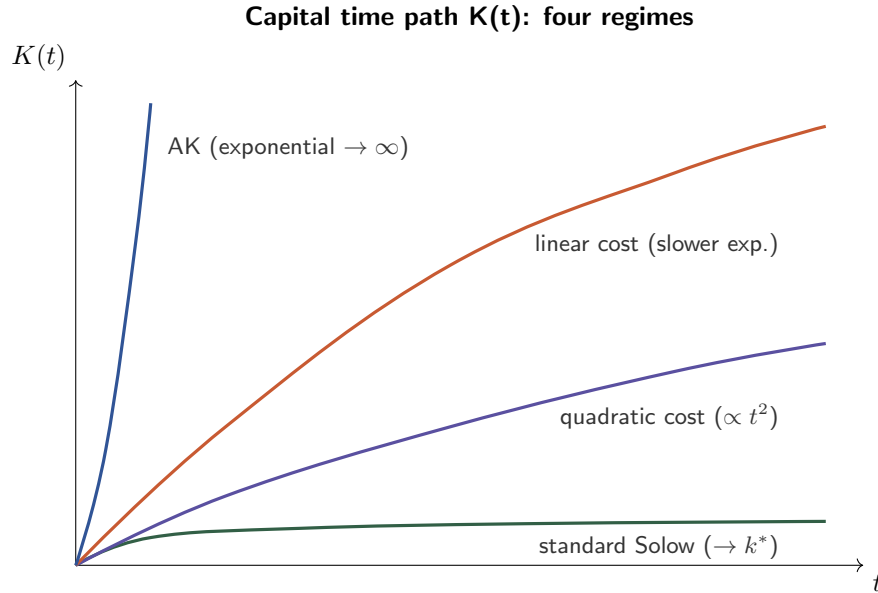


Figure 2: Capital time paths under the four laws of motion (qualitative, theoretical comparison; axes carry no numerical scale). Ordered from fastest to slowest growth: the pure AK model grows exponentially; the linear adjustment cost grows as a slower exponential ($\gamma^\phi < \gamma$); the quadratic adjustment cost grows polynomially ($K_t \propto t^2$); and standard Solow flattens toward its interior steady state k^* . The figure illustrates the qualitative hierarchy of growth regimes, not specific magnitudes.

Anex

We distinguish three objects: the individual firm's capital k_t^j , the aggregate stock K_t , and per-worker capital $k_t \equiv K_t/N_t$. With J identical firms, market clearing requires

$$K_t = \sum_{j=1}^J k_t^j = J k, \quad N_t = \sum_{j=1}^J h_t^j = J h, \quad (44)$$

so the firm's own stock is $k = K_t/J \neq K_t$ in general. Aggregating output, and using that the externality term $K_t^{1-\theta}$ is common across firms,

$$Y_t = \sum_{j=1}^J A_t (k)^\theta (h)^{1-\theta} K_t^{1-\theta} = A_t K_t^{1-\theta} J k^\theta h^{1-\theta}. \quad (45)$$

Substituting $k = K_t/J$ and $h = N_t/J$ and using constant private returns ($\theta + (1 - \theta) = 1$),

$$J k^\theta h^{1-\theta} = J \frac{K_t^\theta N_t^{1-\theta}}{J^{\theta+(1-\theta)}} = K_t^\theta N_t^{1-\theta}, \quad (46)$$

so the number of firms cancels and

$$Y_t = A_t K_t^{1-\theta} K_t^\theta N_t^{1-\theta} = A_t K_t N_t^{1-\theta}. \quad (47)$$